# Raghavan 2021 PDAC — analysis summary

RUN 1 OF 5

## GO / NO-GO GATES

### Gate 1 — mixing & continuity → GO

20/20 PDAC samples (≥120 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 50 cells / 1 speck removed; iLISI median 1.00→7.22 of 20); both lobes (Ductal-body 6,494 / Cycling 674) carry the trajectory. One PANET sample (662 cells, a distinct neuroendocrine malignancy) was excluded to match the declared PDAC scope.

### Gate 2 — module geography → GO (non-branching)

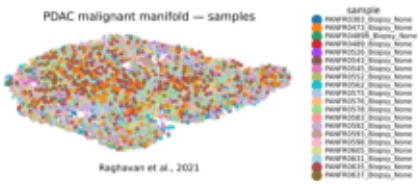
Single dominant pSom→Crest axis: pSom high over the classical-ductal body declines along pseudotime (ρ=−0.74); Crest anti-correlated (r=−0.68), depleted at the root and rising to a distal plateau (ρ=+0.61; crest-sm peaks mid-to-distal, decile pt=0.59). Gast co-varies with pSom (joint-high enrichment 1.15); joint pSom&Crest-high depleted (0.23) — an inverse gradient, not a hub.

### Gate 3 — hub mutual exclusivity → NA

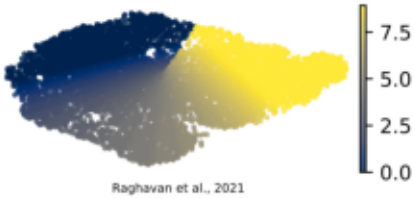
Single monotone axis — no bifurcation and no Mixed-type hub (curve: 2 tips, 0 forks, self-approach 2.57), so there is no branch pair to test for mutual exclusivity.

## STEP FIGURES

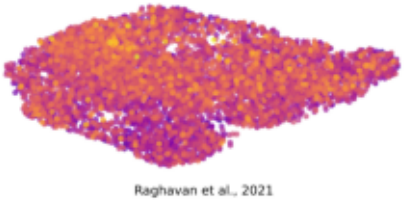
### Manifold — samples



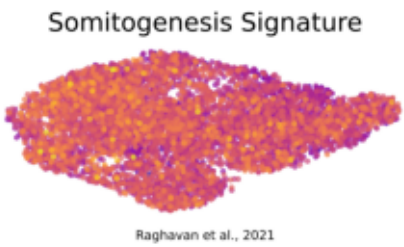

### Pseudotime



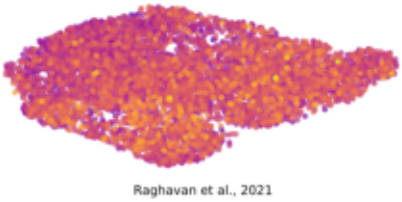

### Gastrulation



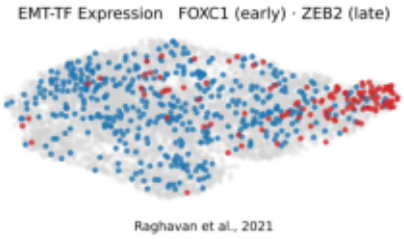

### Somitogenesis




### Neural Crest




### EMT-TF expression




---

**Dataset.**
PDAC (Raghavan et al., 2021), Seq-Well — 24 patients / 24,412 cells (master scope); malignant only: 7,904 cells, of which 7,242 are PDAC (662 PANET neuroendocrine cells dropped). After the ≥120-cell floor, 7,168 cells across 20 samples enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

**Verdict.**
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Raghavan's metastatic PDAC malignant manifold is a single monotone somitogenesis → distal-neural-crest gradient (classical-ductal body → neural-crest-like distal pole); it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both lobes and is orthogonal to cell cycle (module–cyc |ρ|≤0.20; pt–cyc ρ=+0.11), so the whole manifold is kept. Raw module means are pSom-dominant and geography is a coherent inverse pSom→Crest gradient (Crest depleted at root, rising to a distal plateau) — not the mutually-exclusive territories of the failed Peng 2019 PDAC run. EMT-TFs FOXC1 (early/root) → ZEB2 (late) on the widened FOXC/PRRX panel.

## Parameter choices

| | | | |
|---|---|---|---|
| **Min cells/sample** | ≥73 (geom-mean 120→45) | **Scope** | PDAC only; whole manifold |
| **Root node** | tip 0 = pSom-high, Crest-low | **Integration** | Harmony θ=10; PCA50; spread=1.3 |
| **Trajectory** | scFates curve N=20 λ=0.3 μ=100 | **Compartments** | Ductal-body 6,494 / Cycling 674 |